

## Worked Answer Key

### Page 1

1. A line extends forever in both directions; a segment has two endpoints; a ray has one endpoint and extends forever in one direction. Examples: line  $\overleftrightarrow{AB}$ , segment  $\overline{CD}$ , ray  $\overrightarrow{EF}$ .
2. The segment has endpoints  $A, B$ ; the ray starts at  $C$  and passes through  $D$ ; the line has arrows in both directions through  $E, F$ .

### Page 2

1. Parallel lines never meet and stay the same distance apart. Perpendicular lines meet at a  $90^\circ$  angle.
2. They are parallel because opposite sides of a rectangle run in the same direction; they are not perpendicular to each other.

### Page 3

1.  $35^\circ$  acute;  $90^\circ$  right;  $125^\circ$  obtuse;  $180^\circ$  straight.
2. Drawings should have the four stated measures and matching classifications.

### Page 4

1. A correct protractor construction has two rays with vertex at the center and measure  $65^\circ$ ; it is acute.
2. A correct construction measures  $140^\circ$ ; it is obtuse because it is between  $90^\circ$  and  $180^\circ$ .

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1.  $180 - 72 = 108^\circ$ .
2.  $90 - 37 = 53^\circ$ .

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1. Equilateral: all three sides are equal.
2. Isosceles: exactly two sides are equal (5 in and 5 in).

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1. Right triangle: one angle is  $90^\circ$ .
2. Acute triangle: all three angles are less than  $90^\circ$ .

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1. Square; it has four equal sides and four right angles.
2. Trapezoid; draw one pair of parallel sides and one pair that is not parallel.

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1. Heptagon; it has 7 vertices.
2. Triangle—3; pentagon—5; hexagon—6; octagon—8.

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1.  $P = 2(12 + 7) = 38$  cm.
2.  $P = 5 \times 9 = 45$  m.

**Page 11**

1.  $A = 14 \times 6 = 84$  square feet.
2.  $\text{Area} = 11 \times 11 = 121$  square inches;  $\text{perimeter} = 4 \times 11 = 44$  inches.

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1.  $8 \times 3 + 4 \times 5 = 24 + 20 = 44$  square meters.
2.  $10 \times 9 - 3 \times 4 = 90 - 12 = 78$  square units.

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1. Plot the points.  $A$  and  $B$  share a horizontal line because both have  $y = 4$ .
2.  $(5, 3)$ : the first coordinate is 5 right; the second is 3 up.

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1. Both points have  $y = 2$ ;  $7 - 1 = 6$  units.
2. The shape is a rectangle. Side lengths are 4 and 5, so  $P = 2(4 + 5) = 18$  units.

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1. A square has 4 lines of symmetry: two through opposite vertices and two through opposite side midpoints.
2. A non-square rectangle has 2 lines of symmetry, through the midpoints of opposite sides.

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1. An equilateral triangle has 3 lines of symmetry, one from each vertex to the midpoint of the opposite side.
2. A regular pentagon has 5 lines of symmetry, one through each vertex and the midpoint of the opposite side.

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1.  $(3, 2)$  reflected across the  $y$ -axis is  $(-3, 2)$ : the  $x$ -coordinate changes sign and  $y$  stays the same.
2.  $(4, 1)$  reflected across the  $x$ -axis is  $(4, -1)$ : the  $y$ -coordinate changes sign and  $x$  stays the same.

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1. No. It is a rhombus (assuming the sides form a simple quadrilateral); a square must also have four right angles.
2. It is a rectangle. It could be a square only if all four sides were also equal.

**Page 19**

1. Perimeter  $= 2(18 + 9) = 54$  m. Area  $= 18 \times 9 = 162$  square meters.
2. A regular hexagon has 6 equal sides:  $48 \div 6 = 8$  cm.

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1. It is a right angle. Two perpendicular lines intersect to form four right angles.
2. Area  $= 8 \times 5 = 40$  square units; perimeter  $= 2(8 + 5) = 26$  units; a non-square rectangle has 2 lines of symmetry.